

Multiple choice (each question has only one correct answer). You don't need to show your work.

1. (3 points) Let $\vec{v}$ and $\vec{w}$ be vectors in $\mathbb{R}^3$ with $\vec{v} \bullet \vec{w} \geq 4$. Which of the following must be true?

- a) $\|\vec{v}\| \geq 2$ and $\|\vec{w}\| \geq 2$.
- b) The angle between $\vec{v}$ and $\vec{w}$ is less than 90° .
- c) Both $\vec{v}$ and $\vec{w}$ have positive first components.
- d) All of the above.

2. (3 points) Let $P(w, s, m)$ be the yearly profit in dollars to a company when it sells w widgets per year, pays each employee s dollars per year, and spends m dollars per year on materials. Which of the following is a valid interpretation of

$$\frac{\partial P}{\partial m}(3000, 100000, 50000) = -200?$$

- a) If the company makes 3000 widgets per year, pays each worker \$100000 per year, and spends \$50000 per year in materials, then each additional dollar it spends on materials per year results in approximately \$200 less yearly profit.
- b) If the company makes 3000 widgets per year, pays each worker \$100000 per year, and spends \$50000 per year in materials, then its profit is decreasing by \$200 per year.
- c) If the company's yearly profit is going down by approximately \$200 per additional dollar it spends on materials per year, then it is making 3000 widgets per year, paying each worker \$100000 per year, and spends \$50000 per year in materials.
- d) If the company's profit is decreasing by \$200 per year, then it is making 3000 widgets per year, paying each worker \$100000 per year, and spends \$50000 per year in materials.

3. (3 points) Let $f(x, y)$ be a differentiable function that is defined for all $(x, y) \in \mathbb{R}^2$. On which set is f guaranteed to attain a global maximum and a global minimum?

- a) $\{(x, y) \in \mathbb{R}^2 \mid x^4 + y^4 \leq 3\}$.
- b) $\{(x, y) \in \mathbb{R}^2 \mid |x - y| \leq 3\}$
- c) $\{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 < 3\}$
- d) None of the above.

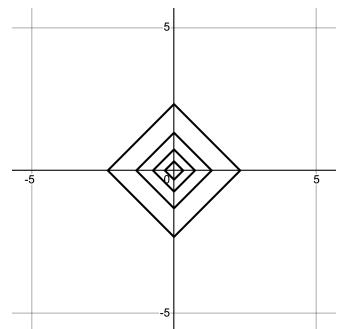
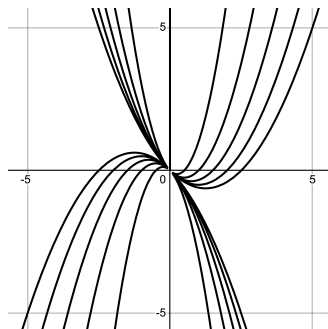
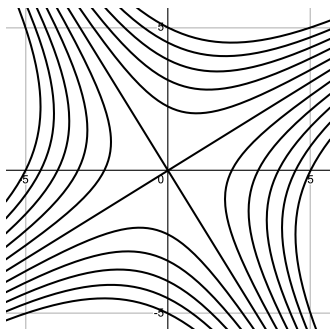
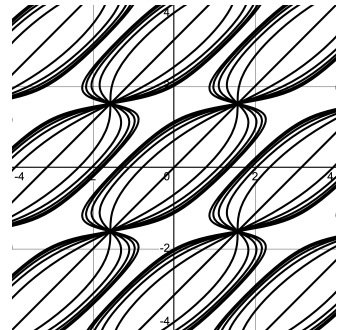
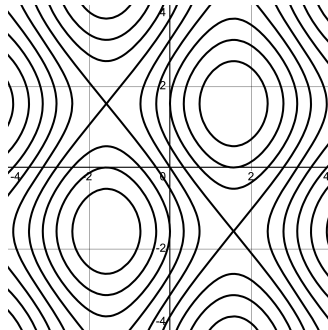
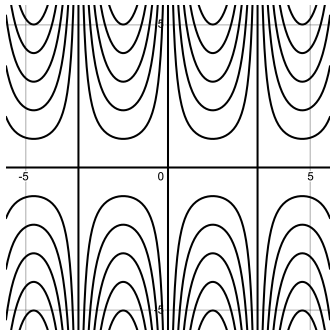
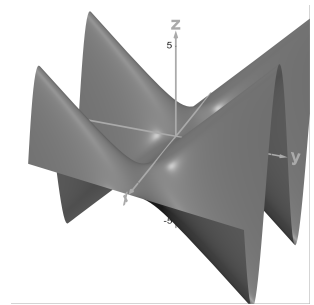
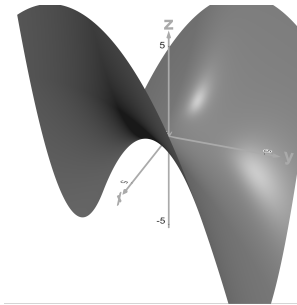
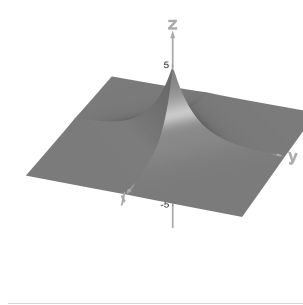
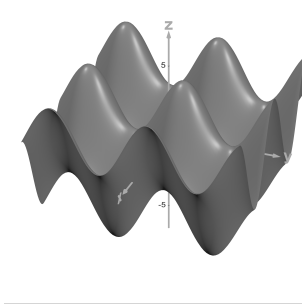
4. (6 points) For each of the following equations, write its letter below its 3D plot and its level curves. Some of the level curves will not be matched.

(A) $f(x, y) = \frac{1}{5}(x^2 - y^2 - 2xy)$

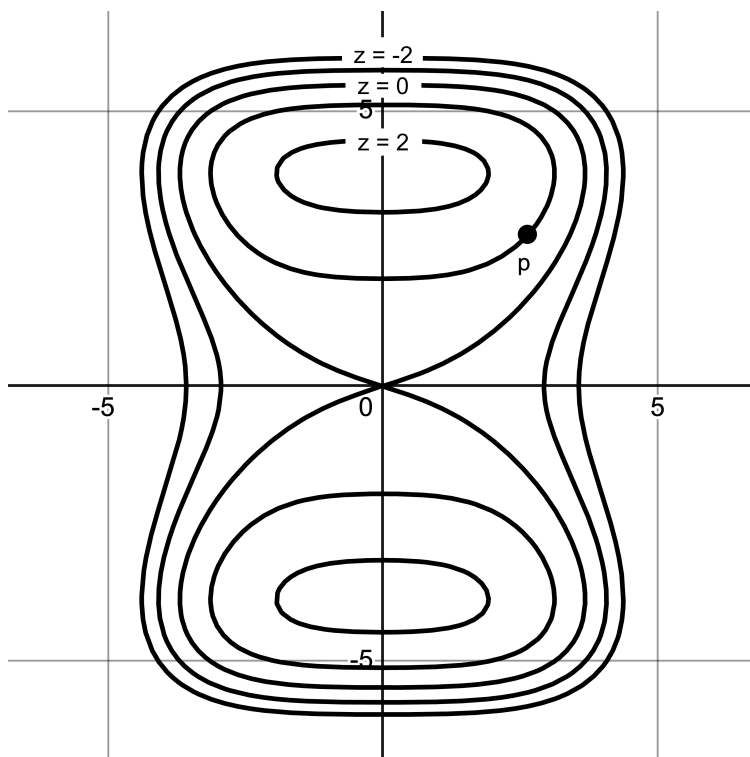
(B) $g(x, y) = y \sin(x)$

(C) $h(x, y) = 5(2^{-|x| - |y|})$

(D) $k(x, y) = 3\sin(x) + 2\sin(y)$



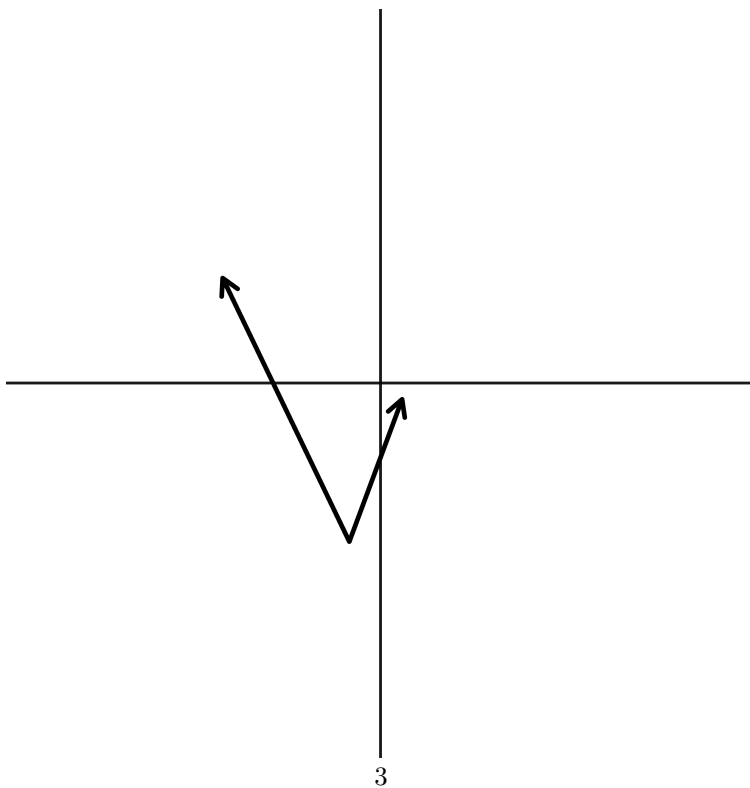
5. (2 points) Below is a plot of level curves of a function $f(x, y)$. Draw a vector starting from the point p in the same direction as $\nabla f(p)$.



6. (4 points) Below are vectors $\vec{v}$ (on the left) and $\vec{w}$ (on the right). Both are in the xy -plane.

a) (2 points) Draw $\vec{v} + \vec{w}$.

b) (2 points) Draw a vector $\vec{a}$ in the plane so that $\vec{v} \times \vec{a}$ sticks up orthogonally out of the page, and $\|\vec{v} \times \vec{a}\| = \|\vec{v}\|^2$.



Longer questions. Show all your work!

7. (12 points) Let $f(x, y) = e^x \sin\left(\frac{\pi}{2}(x + y^2)\right)$.

a) (6 points) Find $\frac{\partial^2 f}{\partial y \partial x}(0, 1)$.

b) (6 points) Find $D_{\vec{u}}f(0, 1)$ for $\vec{u} = \langle 3, 4 \rangle$.

8. (5 points) Let C be the solid cylinder in $\mathbb{R}^3$ with radius 4 that is parallel to the y -axis, and whose center axis passes through $(1, 2, -3)$. Write C in set builder notation.

9. (6 points) The number of hours it takes a student to complete a weekly homework assignment is a function $T(n, h, c)$ of the number of problems n on the homework, the number of hours h that they spend in class and in office hours that week, and the number of friends c that they collaborate with.

Moreover, $c = c(w)$ is a function of the week w of the semester, $h = h(p)$ is a function of the pages of material p covered that week, and $n = n(w, p)$ is a function of both w and p . Given the following derivatives for a particular student at a point in the semester, how many additional hours of homework should this student expect per additional page of material covered?

$$\frac{\partial T}{\partial n} = 1 \quad \frac{\partial T}{\partial h} = -\frac{1}{2} \quad \frac{\partial T}{\partial c} = -2 \quad \frac{dc}{dw} = -\frac{1}{3} \quad \frac{dh}{dp} = \frac{1}{2} \quad \frac{\partial n}{\partial w} = -\frac{1}{2} \quad \frac{\partial n}{\partial p} = 2$$

10. (12 points) Let $p = (1, 0, -6)$, $q = (0, 2, -2)$, and $r = (3, 0, -2)$.

a) (6 points) There is only one plane that passes through p , q , and r . Find an equation for it.

b) (6 points) The line $l(t)$ is parallel to the vector $\langle 1, 2, -1 \rangle$ and passes through the point $(0, -1, 7)$. Find the intersection of $l(t)$ with the plane from part a) or show the two do not intersect.

11. (12 points) Let $f(x, y) = 3x^2 + y^3 - 6xy$.

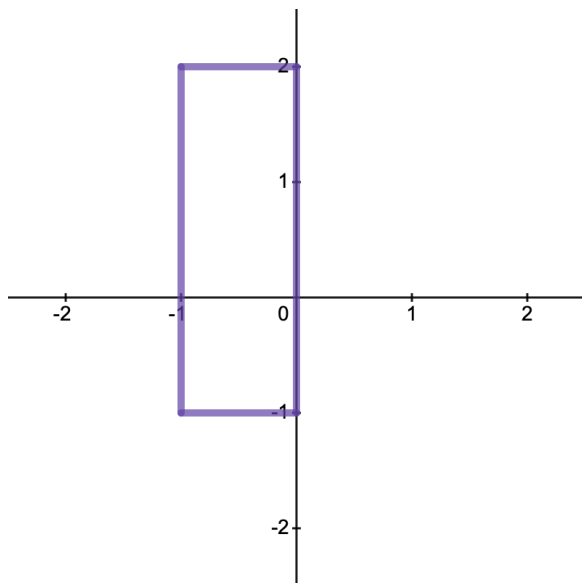
a) (2 points) Find ∇f .

b) (4 points) Find the tangent plane to f at $(1, 1)$ and use it to approximate $f(0, 0)$.

c) (3 points) Find the critical points of f .

d) (3 points) Classify the critical points of f as local maxima, local minima, or saddle points.

12. (8 points) Let $g(x, y) = x^2 + y^4 + 4xy$ be defined on the rectangle R with edges at $x = -1$, $x = 0$, $y = -1$, and $y = 2$, as shown.



Find the maximum and minimum values of g on R . Note that R is hollow: you do not need to consider g inside of it.